

TPM Academy — Capstones

The academy has **two capstone moments**: a mid-arc PRD discipline check in Week 3, and a full-arc artifact-set delivery in Week 8.

	Week 3 — Mini-Capstone	Week 8 — Final Capstone
Position	End of the requirements arc	End of the academy
Subject	FieldPulse (the shared HVAC dispatch SaaS case)	Holocron — an enterprise string-management problem, supplied as a problem brief
Duration	1 week (Mon–Fri)	1 week (Mon–Thu build, Fri present)
AI tools	Off — honor system	On with validation discipline; new AI Spec Development pattern introduced Day 1
Deliverable	One Technical PRD + review-resolution log	6-artifact set (PRD/TCD/TMD/SEP/DP + AI Spec)
Compression rule	None — full 11-section PRD	2 pages per artifact — discipline is what to cut
Triads	Same triads since Week 1	Same triads since Week 1
Friday	Cross-triad peer review (2 primary reviewers + 1 secondary, then revise)	15-min triad presentations + 5 min Q&A; cohort + instructors score
Output proves	Cohort can write a defensible PRD without AI	Muscles from Weeks 1–7 applied end-to-end on an unfamiliar enterprise problem

Week 3 — Mini-Capstone: PRD Discipline

The first capstone is **focused and narrow**: produce one Technical PRD against the shared FieldPulse case, by hand, under peer review. AI is intentionally off all week so the **structured-thinking-expressed-as-written-words** muscle gets isolated and built.

What ships Friday

- A revised PRD (template below)
- A **review-resolution log** showing which feedback was adopted, deferred, or rejected, with reasoning

PRD template (sections used all week, cumulative)

1. Context
2. Problem
3. Goals & non-goals
4. Scope (in / out)
5. Solution sketch
6. Acceptance Criteria (Given/When/Then – granular, falsifiable)
7. Non-Functional Requirements (5 categories)
8. Metrics & validation (tied to Week 2 Tier Sheet)
9. Risks & open questions
10. Dependencies
11. Out-of-scope follow-ups

Daily map

Day	Topic	PRD section produced
Mon	Drafting Technical PRDs	Sections 1–3 (context, problem, goals, scope)
Tue	Granular Acceptance Criteria	8–12 testable ACs
Wed	Non-Functional Requirements	NFR across performance / security / accessibility / observability / compliance
Thu	Mini-capstone PRD assembly	Complete PRD ready for review
Fri	Peer review → revisions → secondary review → sign-off	Revised PRD + resolution log

Review rubric (Friday)

Dimension	Weight
Problem clarity	20%
AC testability	25%
NFR completeness	20%
Strategy linkage (to Week 2 Tier Sheet / journey friction)	15%
Risk honesty	10%
Writing discipline (no AI-generic prose)	10%

The non-AI rule

Allowed: spell-check, dictionary, customer-interview transcripts, Week 1–2 artifacts. Not allowed: any generative AI for drafting, summarizing, or critiquing any part of the PRD.

The reasoning: AI fills vagueness with confident generic prose, and vagueness is the bug Week 3 is trying to eliminate.

Full details: `../../../../week-03/markdown/labs/README.md` . Day-by-day lab briefs: `../../../../week-03/markdown/labs/day-01..05-*.md` . Decks: `../../../../week-03/markdown/presentations/day-01..05-*.md` .

Week 8 — Final Capstone: Integrated Artifact Set

The final capstone is **wide and integrative**. Every triad works the **Holocron** problem (brief issued end of Week 7) and produces a compressed-but-complete version of the 5-week artifact set, plus a new AI-generated integrated technical spec. Nothing is pre-specified — the brief gives business context and facts; the requirements are the triad's work.

What ships Friday

```
capstone/
├── PRD-light.md      ← compressed from Week 3 muscle
├── TCD-light.md      ← compressed from Week 4 muscle
├── TMD-light.md      ← compressed from Week 5 muscle
├── SEP-light.md      ← compressed from Week 6 muscle
├── DP-light.md       ← compressed from Week 7 muscle
├── AI-Spec.md        ← NEW THIS WEEK – integrated engineering-ready
spec
```

Compression constraint: every artifact fits on **2 pages or less**. The discipline is what to cut, not what to add.

Daily map

Day	Topic	Artifact produced
Mon	AI Spec Development technique introduced	AI Spec template + Holocron scope slice locked
Tue	Capstone discovery + compressed PRD	PRD-light
Wed	Capstone architecture + AI Spec drafted	TCD-light + TMD-light + AI Spec v1
Thu	Technical logic validation + finalization	Validated, integrated capstone set
Fri	Final presentations + course closure	Presented capstone; course complete

Friday — presentation day

For ~6 triads: 15 min present + 5 min Q&A each, two morning blocks + two afternoon blocks, then cohort retrospective and post-assessments + course closure.

Presentation rubric

Dimension	Weight
Customer + problem clarity	15%
Integrated artifact set (consistent cross-references)	15%
AI Spec quality (engineering-ready; provenance log)	20%
Trade-off honesty (≥ 2 named with cost + revisit trigger)	15%
Outcome thinking (NS / Tier-Sheet / leading-indicator vocabulary)	10%
Delivery readiness (DP-light: backlog + tracking + one experiment)	10%
Presentation craft (all triad voices; time-managed; curious Q&A)	15%

AI Spec Development (the new technique)

Day 1 introduces a structured prompt sequence that produces an engineering-ready integrated technical spec from PRD/TCD/TMD outputs, with explicit validation discipline. Day 3 applies it. The provenance log — prompts used, what was kept, what was discarded — is required and graded.

Full details: `.././././week-08/markdown/labs/README.md` . Day-by-day lab briefs: `.././././week-08/markdown/labs/day-01..05-*.md` . Decks: `.././././week-08/markdown/presentations/day-01..05-*.md` .

How the two capstones relate

Week 3 proves the cohort can write **one well-formed PRD without AI**. Week 8 proves the cohort can produce the **full artifact set on an unfamiliar enterprise problem in five days, with AI used as a force multiplier instead of a crutch**. The Week 3 PRD muscle is the load-bearing input for the Week 8 PRD-light; every other compressed artifact rests on it.

Failing Week 3 makes Week 8 fragile. Passing Week 3 makes Week 8 an integration test rather than a relearning exercise.